State Commission on Rail Accident Investigation
Ministry of Infrastructure

Report No. PKBWK/03/2018

(summary)

from an investigation of serious accident - category A04
which occurred on 30 August 2017. at 21:53
at Smętowo station, in station track No. 2,
in km 457,485 railway line No. 131 Chorzów Batory - Tczew

area of the infrastructure manager of PKP Polskie Linie Kolejowe S.A.,

Department of Railway Lines in Gdynia

Report approved by Resolution

of the **S**tate **C**ommission for **R**ailway **A**ccident **I**nvestigation

No. 5/PKBWK/2018 of 30.07.2018.

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I.3. Description of the immediate cause of a serious accident and indirect causes established in the proceedings

Based on an analysis of the facts related to the occurrence of a serious accident category A04 on 30/08/2017 at 21:53 at Smętowo station, at track No. 2, at km 457,485 railway line No. 131 Chorzów Batory - Tczew, the research team pointed out the following causes of the serious accident:

I.3.1. Immediate cause

The freight train TMS 564024 failed to stop before the L^2 signposting semaphore, giving the signal S1 "STOP" and entered the intersection No. 24, which was in the way of the train no. MPE 54170.

I.3.2. Primary causes

Driver's insufficient observation of the route of the TMS 564024 train while driving on the secondary main track No. 32 in the Smętowo station, resulting in driver's lack of reaction to the transmitted "Stop" signal on the L^2 semaphore, resulting in entry into gauge track No. 2 at the No. 24 intersection.

I.3.3. Indirect causes

Switching off the train's SHP and CA alertness devices on the S200-303 locomotive; by closing the pneumatic valves of these devices in the pneumatic cabinet of the locomotive, which significantly contributed to reducing the driver's vigilance while driving the train.

I.3.4. Systemic causes

Failure to assess and evaluate risk in the Safety Management System in the scope of the safety routes used after the speed increase to 120 km / h through the station on the main primary tracks, in the situation where switches No. 24 and 25 take part in the course of z^2_{31} and z^2_{32}, which are located in sections of the safety road, and are not dependent on these tracks. The internal regulations of the manager do not specify such situations, in particular when using a multi-block automatic line blockade on adjacent routes.

Justification for the individual causes of the accident in the area of non-compliance the current legal status is given in chapters **III** and **IV** of this Report, describing in detail the course of the event.

I.4. Incident category according to the findings of the investigation committee

Serious accident category – A04

II.1. Identification of a serious accident

II.1.1. Date, exact time and location of the serious accident (station, relation, relation km., route)

Serious rail accident of category **A04** occurred on 30 August 2017 at 21:53 in Smętowo station, in track No. 2, in km 457.485, railway line No. 131: Chorzów Batory - Tczew, area of the infrastructure manager of PKP PLK S.A. Department of Railway Lines in Gdynia.

II.1.2. Description of a serious accident

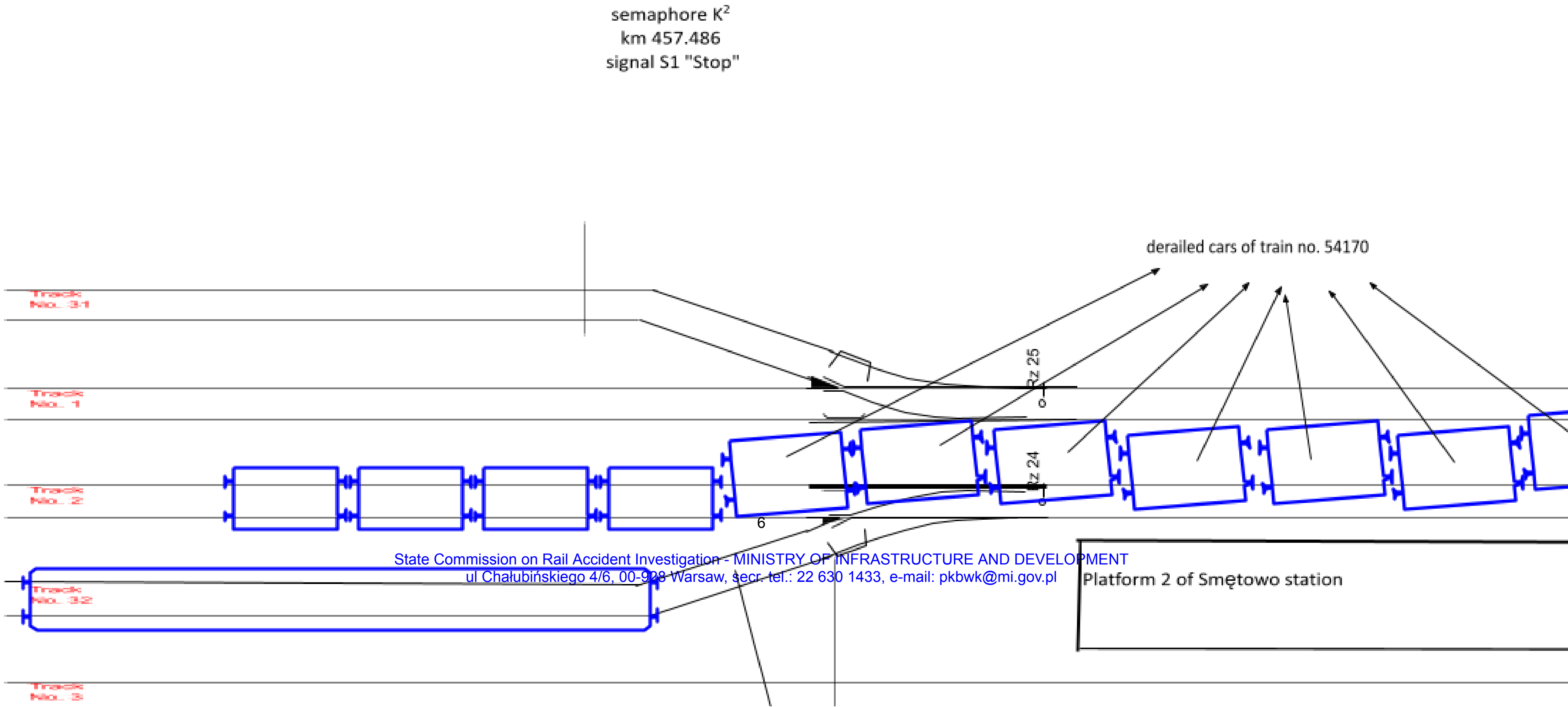
On 30.08.2017 at 21:40, freight train no. TMS
564024– belonging to STK S.A. Wrocław, driven by a driver, left the Morzeszyn station. The train was composed of the S200-303 series diesel locomotive
and 6 cars. At 21:48, the passenger train MPE 54170 of the PKP Intercity SA drove away from the Morzeszczyn station towards the Smętowo station, which had to pass through the Smętowo station without stopping on the station track No. 2, led by the driver with the conductor team, (train manager and conductor).

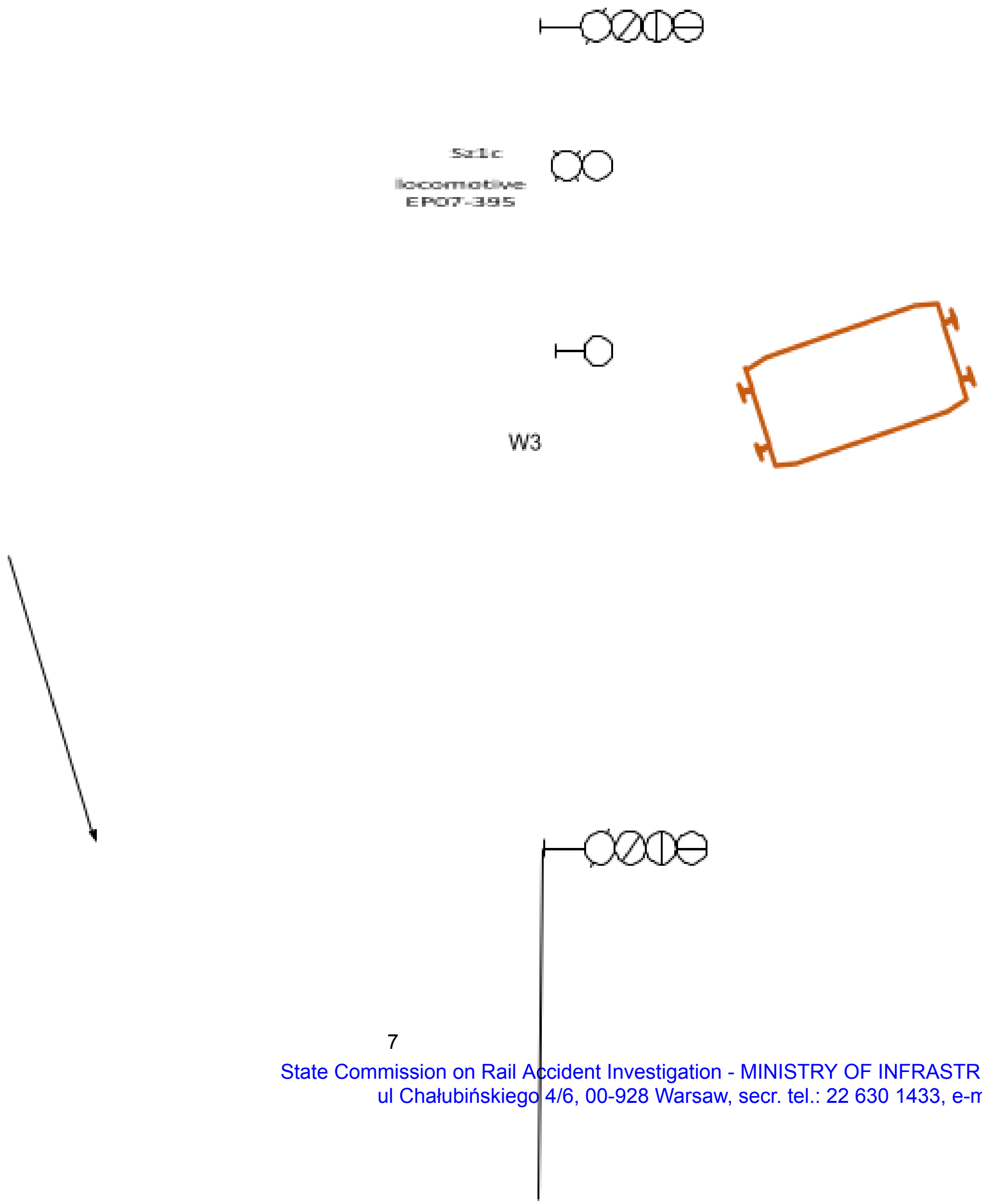
The train was composed of a locomotive and 11 cars: Locomotive EP07 - 395 PL-PKPIC 915111400882. The total weight of the train: 486 tons, length 286 m., percentage of the braking mass required: 104, percentage of the actual braking mass: 141 has been duplicated. Both trains were located on the Morzeszczyn - Smętowo route and moved on the basis of transmitted signals of multiple linear self-blocking semaphores - type Eac. At the time from the opposite direction from the station of Twarda Góra,
track No. 1 to the Smętowo station, the TDE freight train no. 752009 (carrier PKP CARGO S.A.) was approaching, which had to pass through the Smętowo station without stopping on the station track No. 1. Traffic officer and station adjusters at Smętowo station in accordance with the Technical Regulations of the Traffic Outpost and the applicable instructions of the administrator of PKP PLK S.A. prepared the path of the course, giving open signals at semaphores:
- $A^{1/2}$ - signal S2 (one continuous green light), X - signal S2 (one continuous green light) for train TDE 752009 for a drive without stop on track No. 1;
- $Z^{1/2}$ signal S13 (two continuous orange lights) for the TMS train No. 564024 for entry onto the main side track No. 32.

The TMS 564024 train entered Smętowo station on track no. 32 at 21:51 and moved on this track at a uniform speed of about 17 km/h, all the way through. At that time, the MPE 54170 passenger train was approaching the Smętowo station, for which, after preparing the route (drive without stopping), the signal S2 (one continuous green light) was given on the entry semaphore $Z^{1/2}$ and the signal S2 was given on the exit semaphore C (one continuous green light). At the time of signaling, the passenger train MPE 54170 was located at the penultimate distance sbl (fact confirmed by the recording from the video recorder of the locomotive EP07-395). The TMS train 564024 did not stop before the signal S1 indicating "Stop" (one red light) by the signpost semaphore no. L^2 located at track No. 32. The driver of TMS no. 564024 train at the moment when the head of the train was at the height of the L^2 semaphore realized that on the semaphore the signal S1 "Stop" is broadcasted, he triggered emergency braking too late, which caused the train to continue towards intersection No. 24 and the train's head stopped 38m behind the L^2 signpost semaphore. At the same time, the MPE 54170 train was at the station track No. 2. The driver of the MPE 54170 train, having realized that the TMS no. 564024 train is in the gauge of track no. 2, implemented the emergency braking of the train and behind the fouling point of the intersection No. 24 at the speed of 110 km/h there was a clash (right side of the EP07 locomotive) with the left side of locomotive S200 - 303 of the TMS train

No. 564024 located outside the fouling point (indicator W17) at intersection No. 24. As a result of the clash, there was a derailment of the locomotive and seven passenger cars of the MPE 54170 train on the left side in the section between tracks No. 1 and 2. The head of the MPE 54170 stopped at kilometer 457.270 at a distance of 184 meters from the place of the clash. Derailment of the MPE 54170 train caused damage to track No. 1 and its non-occupied control devices, which consequently led to automatic change of image on the entry semaphore $A^{1/2}$ from signal S2 (green continuous) to signal S1 (red continuous) for the TDE freight train No. 752009, which stopped before this semaphore. The station traffic officer at Smętowo, after unsuccessful attempts to call the train drivers MPE 54170 and TMS 564024 on the radio, called the 112 emergency services informing about the railway accident and demanded the arrival of Emergency Medical Services, Fire Brigade and Police. Then he informed the station traffic operators of Morzeszczyn and Twarda Góra not to send any trains to Smętowo. The trains had were correctly marked: front of the train with Pc1 signal (three white lights), the end of the MPE 54170 train the Pc5 signal (two red lights), and the TMS train 564024 with the Pc5 signal (two reflective shields). In order to eliminate the effects of the accident, station tracks No. 1, No. 2, No. 31 and No. 32 were closed. As a result of the accident, 28 passengers of the MPE 54170 train were injured.

Sketch of serious accident category A04 - train No. 564024 failed to stop bef
signpost semaphore L^2, indicating the signal S1 "Stop" and, consequently, c
fouling point
of intersection No. 24 at the Smętowo station with the train no. 54170, on 30/08/




Sz1c
locomotive
EP07-395
W3


locomotive
S200-303 STK
train set
no.
564024
head of the train
no. 564024
km 457.447
semaphore L
km 457.485
signal S1 "Stop"
for train no.
564024

II.1.4. Identification of trains and their trainsets, transported cargo (with particular reference to dangerous goods), railway vehicles, their series and identification numbers, participating in a serious accident, taking into account their previous maintenance..

Train TMS 564024 STK Wrocław S.A. (carrier: STK S.A. in Wrocław), driven by the driver of diesel traction vehicles. Train was composed of
a locomotive and 6 cars. Locomotive S200 - 303 EVN number: PL-ID 92513620076-1. Cars: SK PL-STK 60519908001-7, Rs series PL-STK 33519933008-5, series Rs PL-STK 84513936056-5, Uaai series type Krupp 32 PL-STK 33514940313-0, series Sr PL-STK 37559700002-1, Uaai series, type Krupp 32 PL-STK 83819973320-8

Locomotive type S200 No. 303 serial number 11314 having:

- Certificate no. T / 2003/0095 admission to operation of a railway vehicle type from 28.05.2003. issued at the request of Przedsiębiorstwo Transportu Kolejowego i Gospodarki Kamieniem S.A. in Rybnik by the Main Railway Inspector (unlimited),
- Certificate No. STK / 01/2017 for the technical efficiency of the railway vehicle issued by STK S.A. - on 01/02/2017, expiration date until 31.01.2023 or on the mileage of 600,000 km, counted from 000230 km (at the time of the incident the mileage according to the meter's status was 161 465 km), having the ID of the railway vehicle PL-ID 92 51 3 620 076-1 (S200-303),
- train length: 176 m.,
- overall weight: set/train: 401t. / 516 tons, braking mass required: 200.5 t, which is 50% of the braking mass required, the actual braking mass: 410 tons, which is 102% of the braking mass of the actual train,
- a detailed test of the composite brake: made on 30/08/2017 at the Gdynia Port station, for train no. 564024 by the rolling stock auditor,
- simplified composite brake test: no need to perform.

During the inspection at the scene of the incident and on the basis of data recorded on the tape of the speedometer type **Hasler BERN type RT9 no M08.080** , the commission stated that the driver did not implement "sudden braking" of TMS 564024 train, when the train on track 2 was noticed. The train was controlled from the vehicle's "**A**" console. The train 564024 was stopped, in the fouling point of intersection No. 24 with track No. 2 of Smętowo station, after traveling about 700m from the speed of 17 km/h, bypassing the semaphore L^2 by about 38m.
Siren operation correct. The radio in the train network is set on the channel
no.**6**, consistent with the channel valid on this railway line..
Technical inspections of traction type S200 No. 303: level "**P4**" with modernization - made on 31/01/2017 at DB Cargo Polska SA in Zabrze, level "**P2**" - made on 16/08/2017 by DB Cargo Polska SA – in Rybnik with the meter reading at 11975 km, level **"P1"** - made on 30/08/2017 by a Maxima Spółka Jawna employee
in Gdańsk - with the meter standing at 13590 km, as part of the P1 level review performed by Maxima Sp. J. in Gdańsk, did not check the correctness of SHP pulse alertness recording on the speedometer tape, and did not seal the pneumatic valves of ABP (SHP and CA) Train Equipment control devices in the pneumatic cabinet.

The above results from the scope of activities listed in Part II - Documentation of the S200-303 Maintenance System p. 8 *Inspection and repair sheet - The radio communications system, railway traffic protection and control and measurement devices.*

As part of the review of the "**P2**" level performed on 18/08/2017 by DB Cargo Polska S.A - in Rybnik, a check of the "**Radmor**" radio was carried out, including operation of the "RADIO-STOP" and SHP and active dead-man's handle. **Expertise of the train radio equipment - the F-747 radio showed the lack of the current date and time of the RTC clock of manipulator 0119/OG, which indicates the improper performance of the mandatory review and maintenance of the radiotelephone.**

Analysis of the S200-303 speedometer locomotive strip showed **no registration of the vigilance device resetting signals**, i.e. Self-braking of the Train and Dead-man's Handle (SHP and CA). Expertise carried out in the scope of operation of the **Hasler BERN type RT9 no. M08.080** recorder showed that the device was technically functional. Examination of the reason for the **lack of registration of the vibration warning devices on the speedometer tape (SHP and CA) showed that the reason was switching off the pneumatic valves of the SHP and CA devices in the pneumatic cabinet of the S200-303 locomotive**, which resulted in the display of light and sound signals indicating device's operation from the driver's desk.

MPE **54170** "Pogoria" train from Gdynia Główna - Bielsko Biała Główna, PKP Intercity SA railway carrier composed of the electric locomotive series EP07 No. 395, and
11 passenger cars of the PKP Intercity S.A.

- Certificate No. T/99/0094 of the railway vehicle type approval of 14.12.1999 issued by the Main Railway Inspector in Warsaw for an indefinite period.

- Certificate No. BPT1r-35/2016 regarding technical efficiency issued for the locomotive EP07 No. 395 by "PKP Intercity" SA- in Warsaw on 10.03.2016, expiration date until 15.02.2020 or on the course of 325,915 km, counted from 274 315 km (at the time of the incident, the mileage according to the meter's status was 161 561 km), having the identifier of the railway vehicle PL-PKP IC 95 51 1 140 088-2 (EP07-395)..

Technical inspections of traction type 303E vehicle series EP07 No. 395: level "**P2**" - made on 19.08.2017, by PKP Intercity S.A. - in Warsaw, with the meter reading of 460 820 km, the **„P1"** level - made on 30/08/2017, PKP Intercity SA, the Northern Division
in Gdynia - with the meter reading at 467329 km.

As part of the "**P2**" review performed on 19/08/2017 by PKP Intercity S.A. –
in Warsaw and during the review of the **„P1"** level - made on 30/08/2017,
PKP Intercity S.A. Zakład Północny in Gdynia - the "**Koliber**" radio has been checked, including the operation of the "RADIO-STOP" device and SHP and active dead-man's handle,

- train length: 270 m.,

- overall weight: set/train: 486t./486t., braking mass required: 505t, which is 104% of the braking mass required, the actual braking mass: 690t, which is 141% of the braking mass of the actual train,

- a detailed test of a composite brake: made on 30/08/2017 at 20:15
  in the Gdynia Port station by train 54170 by the rolling stock's auditor,

- 2nd class passenger car No. PL-PKPIC 505184-78001-1 - certificate No. BPT1n-57/2016 of the technical efficiency of the railway vehicle,

-class passenger 2 class No. PL-PKPIC 515120-71046-9 - certificate No. BPT1m-199/2015 technical efficiency of the railway vehicle,

- sleeper car No. PL-PKPIC 525170-80011-0 - certificate No. BPT1a-41/2017 technical efficiency of the railway vehicle,

-class passenger 2 class No. PL-PKPIC 505184-78011-0 - certificate No. BPT1t-13/2017 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 515119-70651-0 - certificate No. BPT1m-34/2016 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 575119-78125-1 - certificate No. BPT1t-59/2015 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 515119-70648-6 - certificate No. BPT1t-124/2016 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 515119-70075-2 - certificate No. BPT1h-91/2017 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 515119-70636-1 - certificate No. BPT1t-138/2015 technical efficiency of the railway vehicle,

-class passenger 1 class No. PL-PKPIC 515119-70084-4 - certificate No. BPT1n-40/2017 technical efficiency of the railway vehicle,

- sleeper car no. PL-PKPIC 525170-80209-0 - certificate no. BPT1p-04/2017 for technical efficiency of the railway vehicle.

All technical efficiency certificates for the above coaches were valid.

II.1.5. Description of the railway infrastructure and signaling system in the place of a major accident - types of tracks, turnouts, srk devices, signaling, SHP, active deadman, including taking into account the current course of their maintenance

Description of Rail Infrastructure:

Railway line no. 131 ……….Chorzów Batory – Tczew

– two-track route …………..........Smętowo – Morzeszczyn,
– route tracks ………………….. No. 1 year of construction – 2011 : No. 2 built– 2012,
– rail type…………………….. 60E1, year of production 2010,
- track base .... track No. 1, prestressed concrete type PS-93 / PS94,
- track sleepers No. 2, prestressed concrete type PS-94,
- fixing .... elastic type SB-3,
- ballast .... crushed stone, 35 cm thick,
- technical condition of the track .... did not affect the occurrence of the event,
- the maximum permitted speed of trains - 160 km / h (fixed in force limitation to 140 km / h is associated with the lack of the required braking distance on the last interval sbl - indicated in the "List of permanent warnings").

Smętowo Railway Station:

Track No. 1 (km 457,107 - 457,460) ... built in 2012
- rails type .... 60E1, built in 2011,
- track base............................. .. wooden oak, built in 2011,
- bolting .... semi-elastic Skl 12 built in in 2012,
- priming .... crushed stone, 35 cm thick.

Track No. 2 from km 457,460 to km 457,107 .... built in 2012
- rails type .... 60E1, built in 2011,
- track base................................ prestressed concrete PS94, built in 2011,
- bolting .... Spring bolts type SB3 built in 2012,
- priming .... crushed stone, 35 cm thick,

- technical condition of the track. .... did not affect the occurrence of the event,
- slope of the track in the direction of travel of the train 0.121 ‰ on the length of 376.00 m,
- the maximum permitted train speed - 120 km / h.

Track No. 32 .... built in 1985
- rails type .... UIC60, year 1985,
- track base ....prestressed concrete bars INBK7, built in 1984,
- bolting....................... ... type K built-in in 1985,
- priming ....crush, thickness 35 cm,
- technical condition of the track. ………………. did not affect the occurrence of the event,
- inclination (slope) of the track in the direction of travel of the train 0.121 ‰,
- the maximum permitted train speed - 40 km / h.

Intersections at the Smętowo railway station taking part in the preparation and fixation of the route:

- intersection No. 49, type Rz UIC60-1: 9-300 lssd, built in 1994,
- intersection No. 46, type Rz UIC60-1: 9-300 pssd, built in 1994,
- intersection No. 45, type Rz UIC60-1: 12-500 pssd, built in 1994,
- intersection No. 44, type Rz UIC60-1: 9-300 pssd, built in 1994,
- intersection No. 25, type Rz UIC60-1: 9-300 pssd, built in 1995,
- intersection No. 24, type Rz UIC60-1: 9-300 lssd, built in 1996.

The technical condition of intersections had no impact on the occurrence of the event.

Railway traffic control devices:

At the Smętowo station, mechanical devices with traffic lights are installed.

Date of construction of devices - 1941.
Reconstruction and modernization of devices:
- addition of traffic lights - 1969,
- adaptation of the Eac type sbl to the four-stationity - 2013.

Execution of routes (road, safety road, side protection) according to
the applicable internal regulations § 37 and 38 "Technical guidelines for the construction of rail traffic control devices" Ie-4 (WTB-E10).

II.2. Fatalities, wounded and losses

II.2.1. Persons injured in an accident, in particular passengers and third parties, railway staff, including contractors

a) Number of people injured in an accident

Category victims	dead	seriously injured	Outpatient assistance or hospital stay up to 24 hours.
passengers	none	10	18
employees in total with subcontractors	none	none	none
railway crossing users	none	none	none
unauthorized persons in the railway area	none	none	none
other…………………………… …	none	none	none

b) restrictions in the movement of trains:

Restrictions in the movement of trains:					
Stoppage in the movement of trains		from time	30.08.2017 21:53	to time	31.08.2017 5:00
delayed passenger trains		number of trains	34	number of minutes of delay	1193
delayed freight trains		number of trains	31	number of minutes of delay	4508
launch of substitute communication		from time	30.08.2017 22:00	to time	31.08.2017 22:35
route closure:	(track) No. 1)	from time:	30.08.2017 22:00	to time	04.09.2017 15:00
	(track) No. 2)	from time	30.08.2017 22:00	to time	04.09.2017 17:15
switching off the voltage in the traction network ..		from time	31.08.2017 02:16	to time	04.09.2017 15:00
re-routing trains		number of trains	none		
shortening the train lines		number of trains	0		
cancellation of trains ……………………		number of trains	0		

II.2.2. Losses arising in cargo, passengers' luggage and other property

In total, passenger claims related to the return of tickets and damaged luggage amounted to 3.9 thousand PLN. There were no losses associated with the transported cargo.

II.2.3. Destruction or damage to railway vehicles, railway infrastructure, environment, etc.

Damages and damages in PKP Intercity S.A. railway vehicles

MPE 54170 "Pogoria" train (KWR-1297615) from Gdynia Postojowa - Bielsko-Biała Główna composed of a locomotive and 11 passenger cars with a total length of 285.9 meters (16.2m locomotive + 269.7m train cars). The weight of the train's cars is 486 tons, the actual braking mass is 690 tons, setting of brakes "R".

The locomotive has been derailed and seven of the eleven passenger cars behind the locomotive. All of the derailed vehicles were located between the railroad tracks 1 and 2, they were recessed in the ground
and tilted to the left in the direction of travel. There is no external signaling of the head of the Pc-1 train due to the lack of power in the locomotive after the incident. Before a serious accident, the head lighting with the Pc-1 signal was working properly. The end of the train is properly signaled by the Pc 5 signal (two red lights).

Locomotive series EP07 - 395 No. EVN PL-PKPIC 91511140088-2, derailed at all four axes. Run from the "B" cab. Serious damage to the chassis (damaged and broken elements of the running gear, vehicle braking system) on both sides. On the box of the locomotive, partial tear in the bottom part of the vehicle with visible damage to the refuge. Significant damage to both walls of the front and bottom parts of the left side of the locomotive box. Damage (denting) to the face of the vehicle's rear cabin has caused a short circuit in the control circuits depriving the locomotive of on-board power supply. The rear pantograph ("A") on which the ride took place was damaged.

Certificate of technical efficiency No. BPT1r-35/2016, issued in Warsaw on 10/03/2016 valid until 15/02/2020, for the course of 325915 km, counted from 274315 km. P4 repair made on 12/02/2014 at BZK Kraków, the last P2 review made on 19.08.2017 in Warsaw at 460820 km, last P1 review made on 30/08/2017 at 13:30 in Gdynia. According to the train record, the accident occurred in the third hour of locomotive operation after the P1 review. Registering speedometer RT9 No. N03.040 indicates the course of 467431 km, the last test of 21/03/2017 (protocol No. 21/17) made at PKP Intercity S.A. Central Department valid until 31.03.2018.

Status of devices and switches in the "B" (leading) cabin:

FV4a driver's valve in full braking position, auxiliary brake valve
in "neutral" position, directional adjuster in forward position, travel adjuster in "0" position, shunt control in position 4, rear pantograph on the console raised, train heating switched on, attached converters and main compressors, three white lights on the head of the locomotive, brake switch in the "fast" position, force adjustment switch
and the high start switch in the off position, sanding on, the speedometer A16 sealed, the speedometer display "0". Radio Koliber KM01 radio manipulator
(No. KM01B3132013) inactive (no power supply).

Status of devices and switches in cabin "A" (inactive):

FV4a driver's valve in cut-off position, auxiliary brake valve in "neutral" position, direction and travel adjusters in "0" position, all switches in off position, brake switch in "fast" position, speedometer readout RT9 16 km / h . Koliber KM01 radio navigator keypad (No. KM01A1222013) is inactive (no power supply). Main cameras SHP and Deadman's Charged Active sealed. Damaged main switch of the on-board battery.

State Commission on Rail Accident Investigation - MINISTRY OF INFRASTRUCTURE AND DEVELOPMENT
ul Chałubińskiego 4/6, 00-928 Warsaw, secr. tel.: 22 630 1433, e-mail: pkbwk@mi.gov.pl

Condition of devices in the machinery space:

Cut-off taps with ABP device switches in the attached position and sealed, status of other valves and switches in the right positions for driving the train.

1) Car 1 - $WLAB^{10}o$ series PL-PKPIC 525170-80209-0

Certificate of technical efficiency No. BPT1p-04/2017 issued on 04.01.2017 in Gdynia, valid until 30.10.2017 for the mileage of 331062 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box, partially torn, the bottom part of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves and on the locomotive.

2) Car 2 - A^9nou series PL-PKPIC 515119-70084-4

Certificate of technical efficiency No. BPT1n-40/2017 issued on 16.03.2017 in Libiszów valid until 15.09.2019 for the mileage of 500,000 km, counted from 180446 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box, partially torn, the bottom part of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves.

3) Car 3 - A^9nouz series PL-PKPIC 515119-70636-1

Certificate of technical efficiency No. BPT1t-138/2015 issued on 08.12.2015 in Libiszów valid until 07.06.2018 for the mileage of 500,000 km, counted from 0 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box, partially torn, the bottom part of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves.

4) Car 4 - A^9nou series PL-PKPIC 515119-70075-2

Certificate of technical efficiency No. BPT1h-91/2017 issued on 21.07.2017 in Warsaw, valid until 30.03.2019 for the mileage of 500000 km, counted from 0 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box, partially torn, the bottom part of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves.

5) Car 5 - A^9nouz series PL-PKPIC 515119-70648-6

Certificate of technical efficiency No. BPT1t-124/2016 issued on 15.07.2016 in Libiszów valid until 14.01.2019 for the mileage of 500,000 km, counted from 83507 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box partially damaged undercoat at the bottom of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves.

6) Car 6 - A^9nouz series PL-PKPIC 575119-78125-1

Certificate of technical efficiency No. BPT1t-59/2015 issued on 28.05.2015 in Libiszów, valid until 27.11.2017 for the mileage of 500,000 km, counted from 8,390 km.

Car derailed with all four axes. Tilted to the left. Serious damage to the chassis (damaged and broken elements of the car running gear) on both sides. On the box partially damaged undercoat at the bottom of the vehicle with visible damage to the refuge. Significant damage to both front walls resulting from the running of the cars on themselves.

7) Car 7 - A^{9}nouz series PL-PKPIC 515119-70651-0

Certificate of technical efficiency no BPT1m-34/2016 issued on 31.03.2016 in Nowy Sącz, valid until 30.09.2012 for the mileage of 500000 km, counted from 4844 km.

Car derailed with all four axles with damage to the chassis. On the box partially damaged undercoat at the bottom of the vehicle with visible damage to the refuge. Damage to the front walls resulting from the running of the cars on themselves.

8) Car 8 - B^{9}nopuvz series PL-PKPIC 505184-78011-0

Certificate of technical efficiency No. BPT1t-13/2017 issued on 07.02.2017 in Libiszów, valid until 06.08.2019 for the mileage of 500,000 km, counted from 165 km.

non-derailed, stopped

9) Car 9 - WLAB^{10}mo series PL-PKPIC 525170-80011-0

Certificate of technical efficiency No. BPT1a-41/2017 issued on 08.08.2017 in Opole, valid until 07.08.2020 for the mileage of 6,500,000 km, counted from 0 km.

non-derailed, stopped

10) Car 10 - B^{10}nouz series PL-PKPIC 51512071046-9

Certificate of technical efficiency No. BPT1m-199/2015 issued on 23.12.2015 in Nowy Sącz, valid until 22.06.2018 for the mileage of 500,000 km, counted from 755,4 km.

non-derailed, stopped

11) Car 11 – B^{8}nopuv series PL-PKPIC 505184-78001-1

Certificate of technical efficiency No. BPT1n-57/2016 issued on 20.05.2016 in Libiszów, valid until 19.11.2018 for the mileage of 500,000 km, counted from 159 km.

non-derailed, stopped

Estimated costs incurred by PKP Intercity S.A. amount to approx. PLN 24.5 million.

Destruction and damage in the traction vehicle S200-303 STK S.A. Wrocław.

Railway vehicles and their equipment:

Initial damage to railway vehicle S200-303 participating in the event:
- broken brake cylinders on the first trolley,
- damaged pneumatic system supplying brake cylinders on 1 trolley,
- damaged shock absorbers on 1 trolley,
- damaged locomotive side support, trolley box with rubber-metal bushings,
- damaged brake system, brake levers, brake inserts,
- deformed vehicle head with maneuvering steps,
- deformed maneuvering platform over one trolley,
- broken door of radiator louvers,
- torn tank cover,
- deformed side skin of the vehicle refuse,
- powdered with fire extinguishing powder: traction engines and vehicle sheeting.

Damage and destruction in the track infrastructure of PKP Polskie Linie Kolejowe S.A.

III.1.1. Organization and manner of issuing and executing commands

PKP Polskie Linie Kolejowe S.A. - infrastructure manager:
This infrastructure manager has: safety authorization:

- EU Number…………………………………………….. PL2120150007
- Date of Issue…………………………………….... 30.12.2015
- Date of validity………………………...………… 30.12.2020
- Type of infrastructure…………………………… normal tracks (99.2%)
  wide tracks (0.8%)
- The size of the managed infrastructure:
  - total length of lines....................... ... ... 18 532 km
  - track length overall .... 36 440 km

- Managed railway lines:
  - magistrate 23%
  - primary 54%
  - secondary 17%
  - local importance .... 6%.

The current "Safety certificate" is an extension of the previous certificate
No. PL2120140003, valid until 29.12.2015.

The manager's Safety Management System was approved by the decision of the President of the Office of Rail Transport No. *TTN-0211-A-07/2010 of 29 December 2010.*

The condition of validity of the decision is full implementation of the rules and conditions for railway traffic safety contained in the document "Safety Management System PKP Polskie Linie Kolejowe S.A.", national and EU legislation as well as continuous fulfillment of the criteria for issuing this document.

The Safety Management System at PKP Polskie Linie Kolejowe S.A., including Zakład Linii kolejowych w Gdyni was introduced by Resolution No. 30/2011 of 24 January 2011 regarding the adoption of an order introducing the Safety Management System at PKP Polskie Linie Kolejowe S.A. adopting Regulation No. 4/2011 of 24 January 2011 of the Management Board of PKP PLK S.A. on the introduction of the "Safety Management System" at PKP Polskie Linie Kolejowe S.A.

Procedure SMS-PG-01 *"Provision of railway lines and operation of railway traffic" version 2.5 of 06.06.2016*

The operation of railway traffic, including railway stations, is described as an element of the main process in the SMS-PG-01 procedure of the Safety Management System (SMS) *"Provision of railway lines and operation of railway traffic".*

In paragraph 1 § 6, the procedure states that railway traffic shall be carried out according to the provisions of the train timetable, instructions, SMS procedures and crisis management procedures.

Details of the permission for entry, transit and departure of the train at the station are specified in Chapter 5 of the Instruction Ir-1 (R-1), 2015 version - consolidated text of 09.05.2017. It states, inter alia:

- § 37 paragraph 1 *"Train movement should be carried out on main tracks, on organized train routes",*
- § 39 paragraph 1 *"Preparation of the route is the responsibility of employees assigned to these activities. If two or more employees on duty at the same time have an on-duty post, it should be included in* 39 par. 2 *"After receiving the command to prepare the route, employees should:*
    a. *remove the rolling stock from the route of the train and interrupt maneuvers (..),*
    b. *make sure, in the manner indicated in the technical regulations, that the route is free from obstacles to the drive, i.e. whether the track is free (...),*
- § 40 paragraph 1 *"Before the signal allowing the train to pass on the semaphore, it is necessary to check whether the route is prepared, i.e. it is properly set and secured and there are no obstacles to the drive",*
- § 41 paragraph 7 *"After receiving notifications about the preparation of the route, provided*
  in § 40 paragraph 6 and after checking this route in their own district, the traffic officer gives the signal on the semaphore allowing the ride or instructs the employee designated by the technical regulations to give this signal (or otherwise allows the train to pass) if all the required conditions are met and the train is approaching or the time of its departure is nearing",
- § 41 paragraph 16 *"The traffic officer on duty is not allowed to give the open signal on entry or signpost semaphore referring to the entry of a train, or to recommend giving this signal, in the case of existence of other obstacles to the drive",*
- § 44 paragraph 1 *"The station employee indicated in the technical regulations, after preparing the route of the train, should observe the setting circle and the passing train until its full entry, departure or transit,*
  and in the event of irregularities threatening the safety of traffic - stop the train or take other actions to eliminate the danger."
  The research team points out that the traffic controller at Smętowo station did not inform the driver of the STK S.A. TMS 564024 train before the train entered the station, about the necessity
  of its transfer to track No. 32, in order to be overtaken by the MPE 54170 train. Such a requirement does not exist in the executive provisions to the Act on railway transport, as well as in the internal regulations of the manager in the field of railway traffic management. The research team is of the opinion that during periodic briefings for traffic officers, it should be noted that this type of information has an impact on the improvement of railway traffic safety and should be provided in particular when the train crosses the traffic post in a way that deviates from the timetable of trains.

<u>Technical regulations for the Smętowo station</u>

To check the way of completing tasks by the traffic controller at Smętowo, the research team analyzed the content of the Smętowo technical regulations - abbreviation "Sm", in force from 31.10.2014. The regulations were approved by the Deputy Director of the Zakład Linii Kolejowych in Gdynia.

Table 1 - Position of the station on the line

Name of the traffic post	No. of line, km	Line category	Number of route tracks	Routes adjacent to traffic post are equipped with devices		
				train operation control device yes/no	No. of train radio channel	Type of block signalling
1	2	3	4	5	6	7
Smętowo	Line no. 131 Chorzów Batory-Tczew km 457.804	magistrate	two	yes	6	automatic, two-way, four-position Type Eac for tracks 1 and 2

Table 2 - Way of managing traffic on adjacent routes

With post (name, abbreviation and type)	Active: (continuously/periodically)	Traffic is run on the basis of:
1	2	3
Twarda Góra Station „TG" announcing post	continuously	Block system: automatic, two-way, four-position Type Eac
Morzeszczyn station „Mo" announcing post	continuously	Block system: automatic, two-way, four-position Type Eac

Table 3 - List of technical posts at the station

Name of post	Abbr.	Crew	Types of railway traffic control devices			Borders of the control range
			station	signaling	line	
1	2	3	4	5	6	7
command box	Sm	One traffic controller	Mechanical	Lights	Block system: automatic, two-way, four-position Type Eac	Line perpendicular to the axis of the track and south face of the building of the former wagon weight at km 457.275 along the track 11 to U12 semaphores at km 74.778 (line 218), T12 at km 457.939 perpendicular to the track axis.
dependent box	Sm-1	One signal operator	Mechanical	Lights	Block system: automatic, two-way, four-position Type Eac	From the A1 / 2 l semaphores, in km 456,495 (line 131), B12 in km 76.250 (line 218) to the southern end wall of the former car weight building at km 457.275 perpendicular to the track axis.
dependent box	Sm-2	One signal operator	mechanical	Lights	Block system: automatic, two-way, four-position Type Eac	From the entry semaphore Z12 in km 458,832 perpendicular to the axis of the track at the height of the T12 semaphore at km 457.939

Table 4 - List of activities that the dispatcher of the control room having "Sm" should perform at the entrance of the train in accordance with the course of Z^1_2 included in the Technical Regulations of the Smętowo Station - plot no. 31

31. Arrival of the train on the permissive signal on the semaphore

Control post: dependent box„Sm-2"

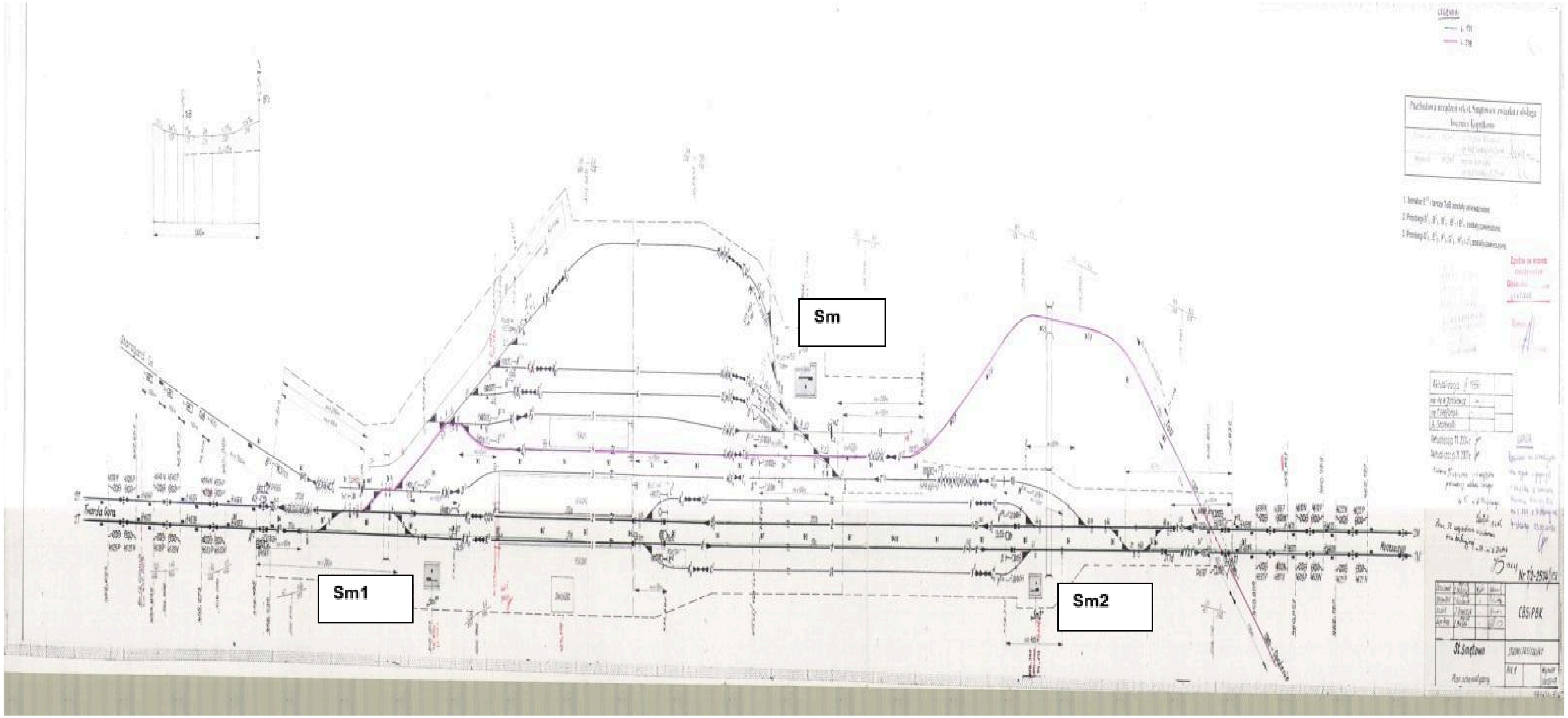
No.	Description of activity:		Execution of activity			
	Route symbol or group of routes		Z^1_2	Z^2_{31}	Z^2_{32}	Z^2_{103}
1	Checking the occupancy of track	Who checks	signal operator Sm-2			
2		Manner and location of checking	Track isolation, visual through an open window of the command post Sm-2			
3		Range of checking	from semaphore T^1_2 to semaphore Z^1_2			
4	Withholding maneuvers during drive	Number of track(s) where maneuvers need to be withheld	2.32	1.31	2, 32	2,32,3,103
5		How many minutes before the drive the maneuvers need to be withheld	10	10	10	10
6		Who gives the order	Traffic controller			
7		Who reports the completion of the order	Maneuver manager of the carrier			
8	Preparing the drive route and reporting on its readiness	Command boxes (posts) involved	Sm Sm-1 and Sm-2			
9		Who controls the drive route	signal operator Sm-2			
10		Who confirms the drive route	signal operator Sm-2			
11		Manner of confirming the readiness of drive route	Switching route lever			
			Z^1_2	Z^2_{31}	Z^2_{32}	Z^2_{103}
12	Location of site:	of signaling	command box axis			
13		of routing	with isolated intersection connection point			
			no.44	no.43	no. 44	no.45
14	Observation of train entry	Who observes	signal operator Sm-2			
15		From which position	through an open window of the command post Sm-2			
16	Operation of the semaphore, block system and station block system	Who operates the semaphore	signal operator Sm-2			
17		The manner of ensuring the provision of "go" signal on the semaphore by the traffic controller	On lights plan - repetition signal			
18		The manner of ensuring the provision of "Stop" signal on the semaphore by the traffic controller	On lights plan - repetition signal			
19		Who operates the block system	-----------------------------			
20		Who operates the station block system	signal operator Sm-2			

* in the conditions of poor visibility and absence of non-occupancy control devices, check on the ground:

After completing maneuvering at the station, the maneuver manager checks on the ground whether there are no brake shoes left on the track.

The fact of checking being complete shall be reported to the traffic controller on duty.

Drawing - Schematic plan of Smętowo station



„PKP Intercity” S.A. – railway carrier:

The above-mentioned railway carrier has:

1) Safety certificate - Part A:

- EU Number…………………………………………….. PL1120150041
- Date of Issue…………………………………….... 15.12.2015
- Date of validity……………………….………… 15.03.2020
- Type of transport .... passenger transport, including rail transport high speed
- Transport volume .... 200 million passenger kilometers per year
- The size of the enterprise .... large

2) safety certificate - part B:

- EU Number…………………………………………….. PL1220150041
- Date of Issue…………………………………….... 31.12.2015
- Date of validity……………………….………… 31.12.2020
- Type of transport .... passenger, including high-speed rail transport
- Supported lines: PKP Polskie Linie Kolejowe S.A. and PKP Szybka Kolej Miejska w Trójmieście Sp. z o. o.

The following procedures are relevant for the Safety Management System of PKP Intercity S.A. :

- ✔ Procedure **P-05** - *"Implementation of the transport process",*
- ✔ Procedure **P-09** - *"Personnel management",*
- ✔ Instruction **Bt-1** - *"Instruction for traction train driver".*

Procedure **P-05** *"Implementation of the transport process"*

The purpose of the **P05** procedure is the preparation and implementation of rail transport in accordance with the current timetable - Appendix A and B and legal regulations and laws issued by the carrier, infrastructure managers and other entities involved in the preparation and implementation of rail transport. The subject of the procedure are all activities related to the preparation and performance of the transport, establishing a uniform procedure for all participants taking part in the preparation process
and carrying transport of people and things on the railway network.

As part of the **P-05** procedure, before starting the train, the carrier shall in particular plan the work of the rolling stock and plan the circulation of the train teams.

Designated employees of the "Zakład" are responsible in particular for providing a train crew with current licences in accordance with the criteria specified in procedure **P-09**, i.e. current exams, medical examinations, periodic training, authorization and knowledge of the route and stations along the route.

In addition, the designated employees of the "Zakład" are also responsible for providing the train crew: timetable, List of permanent warnings, extracts of technical regulations of stations, provision of equipment with written orders and a list of railway vehicles in the train, as well as providing the equipment with a brake test card.

The research team of the PKBWK as a result of the analysis, stated that the carrier, in a manner consistent with the procedure in question, provided the traction team for servicing the train, as well as the operation and equipment of the carrier's locomotive was in accordance with applicable regulations.

Procedure **P-09** *"Personnel Management"*

The purpose of the procedure is to determine the procedure for managing the employees of PKP Intercity S.A. directly related to the operation and safety of rail traffic. The procedure specifies all activities related to the management of employees directly related to the conduct and safety of railway traffic, ranging from employment planning, recruitment, preparation for work, professional development and termination of employment.

The procedure also specifies the conduct in the process of employee management on driver's seat in accordance with the *regulation of the Minister of Infrastructure and Development of 10 February 2014 on the driver's licenses (Dz. U. item 211, as amended)* and the *ordinance of the Minister of Infrastructure and Development of 10 February 2014 regarding the driver's licenses* (Dz. *U. item 211, as amended)*

The research team of the PKBWK as a result of the analysis, stated that the carrier provided a traction team meeting the criteria and requirements referred to in procedure **P-09.**

Instruction **Bt-1** *"Instruction for traction train driver"*

Document related to the application of SMS procedures "PKP Intercity" S.A. No. **P-05 and P-09** is in particular instruction **Bt-1** *"Instruction for traction train driver"*.

This instruction sets out the scope of basic duties and responsibility of the driver of the traction vehicle.

Pursuant to § 13 paragraph 1 point 5 of the Instruction **Bt-1**, the driver is obliged to immediately stop the train, among others if they notice an obstacle on their or an adjacent track that prevents passage. Before the serious accident, the train driver MPE 54170, for about 2 sec. before the occurrence of the event, seeing the locomotive of the TMS train 564024 standing on the right in the gauge of track No. 2 of the Smętowo station, launched a sudden braking procedure, followed by a side collision of both trains and consequent derailment of the locomotive EP07-395 and 7 passenger cars.

The research team after analyzing the recording from the camera recording the image in front of the vehicle and the sound in the cabin, analyzing the train radio record and after listening to the traction team's employees and analyzing the available train documentation and Smętowo station, states that the train driver of the MPE 54170 train, during and after the incident acted correctly and did not contribute to the occurrence of a serious accident.

Picture from about 80 meters before the collision site

View from the PKP Intercity S.A. locomotive cabin from about 40 m before the impact

Locomotive cabin lighting switched on

View from the PKP Intercity S.A. locomotive cabin from about 10 m before the impact

STK S.A. – railway carrier:

The above-mentioned railway carrier has:

1) Safety certificate - Part A:

- EU Number……………………………………………….. PL1120150039
- Date of Issue…………………………………….... 29.11.2015
- Date of validity…………………………..………… 29.11.2020
- Type of transport……………………………… freight, including transport of hazardous goods
- Volume of transport………………………….. 500 million or more tonne-kilometers per year
- The size of the enterprise .... medium

2) Safety certificate - Part B:

- EU Number……………………………………………….. PL1220150046
- Date of Issue…………………………………….... 22.12.2015
- Date of validity…………………………..………… 22.12.2020
- Type of transport - freight, including transport of hazardous goods
- Supported lines: PKP Polskie Linie Kolejowe S.A., PMT Linie Kolejowe Sp. z o. o., CTL Maczki – Bór Sp. z o. o. , Jastrzębska Spółka Kolejowa Sp. z o. o. , Kopalnia Piasku Kotlarnia – Linie Kolejowe Sp. z o. o. Infra Silesia S.A., PKP Linia Hutnicza

Szerokotorowa Sp. z o. o. PKP Szybka Kolej Miejska w Trójmieście Sp. z o. o., Euroterminal Sławków Sp. z o. o.

The following procedures related to the Safety Management System of the STK S.A. :

- ✔ Procedure **P/04** - *"Implementation of the transport process",*
- ✔ Procedure **P/12** - *"Personnel competence management",*
- ✔ Instruction **STK-1**- *"Instruction for driver and helper of diesel and electric locomotive",*
- ✔ Instruction **STK-6** - *"Instruction on professional preparation".*

Procedure **P/04** *"Implementation of the transport process"*

The purpose of the **P / 04** procedure is to present the course of the transport process together with the course of the process
of transporting dangerous goods and extraordinary items by rail. The procedure specifies all activities related to planning
and organizing the transport process setting a uniform procedure for all participants taking part in the process of transporting goods on the railway manager's network (including dangerous goods and extraordinary items). The transport process is the carrier's main process according to the SMS Book.

As part of the transport process, the following activities are performed in particular:

- conclusion of a contract or acceptance of an order,
- planning rolling stock that meets technical, commercial and insurance requirements,
- planning qualified traction service providers,
- determination of the psychophysical state of the railway vehicle crew,
- providing the traction team equipment with the required documents, e.g. written orders, current timetable, WOS, traction teams work card, brake test card,
- supervision of loading,
- inspection and checking the technical condition of cars included in the train,
- inspection, checking the technical condition of the locomotive and documents,
- checking the efficiency and operation of SHP safety devices, active dead man's handle and Radiostop with recording in the on-board book of the vehicle with the drive,
- performing a train brake test,
- drawing up a list of railway vehicles in the train,
- carrying out control braking,
- train running, including adherence to signals and indicators along the route of the train, observation of the route, supervision over the technical condition of railway vehicles and cargo,
- ending work on the locomotive and providing the dispatcher with the work card together with speedometer tape and train and transport documents.

The procedure P/04 is related in particular to the STK-1 and STK-2 instructions.

STK-1 *"Instruction for driver and helper of diesel and electric locomotive",* The instruction specifies the scope of basic duties and responsibilities of the driver of an electric or diesel locomotive employed at STK S.A. servicing traction vehicles while running trains and maneuvering work.

In accordance with the provisions of the Instruction:

1) § 13 *"Train departure may take place after receipt of proper train documents (list of cars in the trains and brake test card) and permit for driving",*
2) § 15 paragraph 2 point 1, the driver is obliged to *"watch closely the signals, strictly follow them and pay attention to the train",*
3) § 15 paragraph 2 point 2, the driver is obliged to *"observe the path of the course while traveling within the station",*
4) § 15 paragraph 9 *"When approaching the semaphore indicating the signal "Stop", the driver should adjust the speed so as to stop the train as close as possible to the semaphore, but without the risk of omitting it, where the semaphore should be visible through the windshield of the traction vehicle",*
5) § 19 paragraph 1 point 6 during ending work on the locomotive to the duties of the driver, it is necessary to "take off and describe the speedometer tape".

STK-2 *"Instructions for the rolling stock auditor"*

The instruction sets out the scope of basic duties and responsibilities of the rolling stock auditor and the employee designated to oversee and coordinate the work of the auditors of the rolling stock of the railway carrier STK S.A.

In accordance with the provisions of the Instruction:

1) § 4 paragraph 1. The auditor's responsibilities include *"making a brake test",*
2) § 8 paragraph 2 point 9 When starting a train, the rolling stock auditor should *"perform the required brake test and fill in the brake test card".*

The research team, as a result of the analysis of the above documents, stated the following irregularities in their application by the employees of the STK S.A. carrier:

1) The driver of TMS 564024 train did not comply with the provisions of procedure P/04 and § 15 paragraph 2 point 1 of the STK-1 Instruction and did not comply with the indications of the L^2 signpost semaphore of station Smętowo indicating signal S1 "Stop", § 15 paragraph 2 point 2 of the Instruction and in an insufficient way watched the course of the course, as well as § 15 paragraph 9 of the Instruction and did not regulate the speed of the train to stop the locomotive's head before the L^2 semaphore indicating the "Stop" signal, which caused passing this semaphore, entering and stopping the head of this train at the gauge of track No. 2, occupied by the MPE 54170 train of PKP Intercity SA, as a result, there was a clash between both trains and, as a consequence, the derailment of the MPE 54170 train.
The research team recognizes the above circumstances as the primary cause of a serious accident.

2) The driver of the TMS 564024 train did not comply with the provisions of procedure P/04 and § 13 of the STK-1 Instructions and, despite the lack of a brake test card in the locomotive cabin S200-303, he left the Gdynia Port GPA start station. The inspection carried out by the PKBWK and the railway commission after the occurrence of the incident showed that the brake test card was in the social car in the train. The research team considers this fact as another irregularity not related to the causes of the event.
3) Carrier's employees employed in the position of train driver did not comply with the provisions of procedure P/04 and § 19 paragraph 1 point 6 of the STK-1 Instruction and after completing the train drive with this locomotive on 29.08.2017 failed to remove the speedometer tape from the recorder
and describe it.
4) Similarly, after the completion of one of the previous locomotive drives, the carrier's employees employed as drivers did not comply with the provisions of procedure P / 04 and
§ 19 paragraph 1 point 6 of the STK-1 Instruction and after finishing the train drive with this locomotive, they failed to describe the speedometer tape. The above statement is confirmed by the fact that the visual inspection carried out by the WCWK and the railway commission after the occurrence of the incident showed in the locomotive STK200-303 had non-delivered and non-described speedometer tapes. These tapes came from servicing trains run by this locomotive in previous days. It is impossible to determine the train numbers and the dates of their journeys. The above also indicates the lack of proper supervision by the carrier over the circulation of speedometer tape archives after finishing the work of the traction teams. The research team considers this fact as another
irregularity unrelated to the causes of the event.
5) The auditor carrying out a detailed test of the TMS 564024 train brake did not comply with the provisions of § 8 paragraph 2 point 9. of Instruction STK-2 and after filling in the brake test card did not give it to the train driver for signature, i.e. did not give the driver one signed brake test card signed by the auditor and driver, and additionally he signed the driver's test name with the driver's name and surname in the place where he should sign train driver, thereby falsifying the train documentation. This fact is confirmed by the hearing of the driver of TMS 564024 train. The research team considers this fact as another irregularity not related to the causes of the event.
6) The driver of TMS 564024 train has not complied with the provisions of procedure P / 04 of the Safety Management System transport procedure for the completion of the "List of railway vehicles in the train set". The list of railway vehicles in the train for the TMS train No. 564024 was drawn up by an unauthorized person - the rolling stock auditor, and should be prepared by the train driver. The research team considers this fact as another irregularity not related to the causes of the event.

Procedure P / 12 *"Personnel competence management"*

The purpose of the procedure is to present the course of the process of competency management of employees related to railway traffic safety.

The procedure specifies all activities related to the management of employees' competences directly related to railway traffic safety, i.e. driver of traction vehicles, driver's assistant, rolling stock auditor, setter and maneuver.

In relation to people employed as a train driver, the procedure divides them into two groups:

- driver of traction vehicles employed according to the old rules in force until 29.10.2018.
- driver of traction vehicles employed according to the new rules in force from 04.03.2014.

The procedure defines the following areas related to employed employees in security-related positions, including train drivers:

1) Employment of staff, in particular:
   - analysis of training needs,
   - recruitment of employees with verification of qualifications and entitlements they possess,
   - application for employment,
   - referring an employee to medical examinations,
   - safety training and in justified cases - training in the safety of dangerous goods transport,
   - employment of an employee - signing a contract of employment.
2) Periodic advices, in particular:
   - drawing up annual programs of periodic instructions, a list of employees subject to instructions and a journal of instructions,
   - choosing an external supplier to conduct instructions or their independent implementation,
   - realization of instructions,
   - conducting a test of knowledge for employees absent from periodic instructions,
   - monitoring the implementation of instructions.
3) Staff examinations - according to the old system (management law):
   - specification of the type of the exam (qualifying, periodic, verification),
   - selection of an examination board appointed by UTK,
   - conducting the exam,
   - issuing documents related to the exam, including the right to manage, exam report, etc.,
   - carrying out authorization, if required.
4) Training and exams of employees - according to the new system (license):
   - determination of training needs,
   - selection of training and examination center,
   - implementation of the training,
   - license exam,
   - issuing a certificate of passing the exam,
   - application to UTK for a license,
   - transfer of the license to the driver after its release by UTK.
5) Training and exams of employees - according to the new system (license):
   - determination of training needs,
   - selection of training and examination center,
   - implementation of the training,
   - license exam,
   - issuing a certificate of passing the exam,
   - application to UTK for a license,
   - transfer of the license to the driver after its release by UTK.

6) Training and exams for train drivers (driver's certificate):
   - determination of training needs,
   - selection of training and examination center,
   - implementation of the training,
   - driver's certificate test,

- handing over the documentation related to the exam to the company by the training center,
- issuing a driver's certificate to the employee by the company.

In the area of employee competencies, the document related to the P / 12 procedure is the STK-6 instruction titled "Instruction on professional preparation", in which the above-mentioned areas of the procedure are described in detail.

The driver was employed on the basis of the contract no. 12/2017 of 27.12.2016.

The research team states that the driver has been employed in a manner inconsistent with the binding one in the Company, procedure P/12, which requires "signing a contract of employment", not contract of mandate.

Register of threats of STK S.A.

As part of the Safety Management System (SMS), the company runs the so-called "Hazard register". The latest version of this document prior to the occurrence of the investigated event was released on 10.05.2017.

The hazard register contains the following elements:

- number,
- type of threat,
- possible consequences,
- risk control measures,
- principles of risk acceptance.

The functional areas of the "Hazard Register", to which the examined event relates, were included by the carrier in Part 6 "Implementation of the transport process", respectively.

The research team states that the events under investigation relate to the hazards described in detail in the following parts of the Register:

- point 6.7 'incorrect observation of the driver's foreground',
- item 6.19 "not stopping the vehicle / train in front of the signaling device indicating signal S1" Stop ",
- point 5.5 "non-compliance with the system procedures by employees", - the carrier's employees did not comply with the provisions of procedure P / 04 and § 19 paragraph 1 point 6 of the STK-1 Instruction and after finishing one of the previous drives of the locomotive they did not describe the speedometer's tape,
- point 6.1 "non-compliance with the instructions in force at PKP PLK SA",
- point 6.4 "unreliable performance of the brake test" - the auditor carrying out a detailed test of the TMS brake number 564024 did not comply with the provisions of § 8 paragraph 2 point 9. of STK-2 Instructions.

The above elements, described in point 6.7 and 6.19 of the Hazard Register, are, according to the Research Team, the primary and indirect cause of the event under examination.

The research team recommends supplementing the STK S.A. Hazard Register with the following aspects identified during the event investigation, i.e.

- lack or incorrect registration of parameters on the strip of the locomotive speedometer,

- disconnecting SHP devices and active deadman's handle by closing the pneumatic valves of these devices,
- not describing the speedometer's tape by the driver and / or not delivering it to the dispatcher after the completed work shift.

The program to improve the safety of STK S.A.

The program for 2017 includes, inter alia:

- conducting periodic instructions in the field of rail and traction traffic for employees directly related to railway traffic safety,
- training in the field of terrorist threats for directly related employees with railway traffic safety,
- control of safe maneuvering,
- testing the sobriety of employees,
- improving the qualifications of railway commission members,
- conducting periodic instructions - theoretical and practical classes.

The research team is of the opinion that the following should be additionally included in the program to improve the safety of the carrier, in particular:

- checks on the work of train drivers,
- controls on the correctness of entries on speedometer tapes or on digital media,
- increased supervision and control of documentation related to the performance of transport, including control of the brake test cards.

III.3. Summary of hearings

The description of the hearings concerns a serious accident cat. **A04**, which took place on 30.08.2017. at. 21:53 at the Smętowo station.

The personal data of the employees heard are subject to protection in accordance with the requirements of Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016.on the protection of individuals with regard to the processing of personal data and on the free movement of such data and the repeal of Directive 95/46 / EC (general regulation on data protection (O. J. EU L 119, 04.05.2016, p. 1, as amended) and related thereto Act of 10 May 2018 on the protection of personal data (Dz. U. item 1000).

Driver of the TMS 564024 train

Hearing of 04.09.2017

The TMS 564024 train driver was trained in the scope of SMS. The training took place at the headquarters of the STK S.A. company in Wrocław during periodic instructions. He does not remember the dates.

1. From the hearings of employees related to the incident, it follows that the train driver 564024 after entering the track no. 32 at Smętowo station did not comply with the signal indications on semaphores and did not stop the train before the L^2 semaphore indicating signal S1 "Stop", being aware that entry to this the track was followed by two orange lights. Departure from this track

should be on signal S10 - "Driving at a speed not exceeding 40km / h, and then with the highest speed allowed"
, not S2 -"Driving with the highest permitted speed", which he thought he should apply to. When he realized that he omitted the semaphore indicating "Stop" he did not implement sudden braking, which was revealed by the speedometer tape
test from the recorder, he only changed the direction of the controller and after a while he implemented a sudden braking. After entering the track no. 32 he wanted to make contact
with the traffic controller, but he did not stop the train. The driver claimed to have pressed the "Radio-Stop" button but the button did not work. The expert opinion of the train radio equipment made by its manufacturer showed that there was no use of the "Radio-Stop" signal. If the semaphore was omitted, the train driver confirmed that his colleague from the train was informed of this fact after the occurrence, which was confirmed by the colleague.

2. The driver of Intercity S.A. train entering the station, seeing on the track on the right the freight train gave the signal "Attention". During passing, he noticed that in the gauge of track No. 2, on which he rode, there is the front of the diesel locomotive of this train. Immediately, he turned the controller to zero and implemented a sudden braking of the train shortly before the impact. He took notice of the semaphore on the right side of the freight train, which indicated the signal "STOP". He wanted to escape to the engine room, but he did not make it because there was a strong impact from the right side of the locomotive. The force of the blow threw the driver from behind the adjuster
and hit the right wall of the locomotive's cabin, and then he threw it to the left. After stopping the train, he got up from the floor and saw the front lights of the train approaching the Smętowo station through the windshield of the locomotive
from the opposite direction. He tried to use the "ALARM" button and illuminate the front of the locomotive, but found no 110V power supply. At the same time, he felt the smell of burning insulation. When the driver found the flashlight he stated that the train from the opposite stopped before the entry semaphore, so he quickly looked for a mobile phone and called the IC dispatcher in Bydgoszcz and informed him about the accident. Then he went into the engine room to find out what had happened and whether there was a locomotive fire. After arriving in cabin A, he stated that there was no fire, only the main switch of the locomotive's battery was burnt. Then he left the locomotive to get an idea consequences
of derailment and possible assistance to victims. Seeing the enormity of destruction and the cars lying on the left side, he headed towards the train depot, where the train manager and the conductor assisted the travelers.
3. The Smętowo station traffic officer and the adjusters properly prepared the route of the train for both trains, which was confirmed in the records of the documentation kept at the checkpoints as well as in the telephone conversations.
4. During the hearing, the drivers of locomotive S200-303 did not confirm any related faults with the operation of ABP devices
on the locomotive, although the evidence in the form of disabled pneumatic valves of these devices on the locomotive and conducted inspection of the vehicle series S200 clearly showed that the reason for the lack of registration on the speedometer tape parameters related to the operation of SHP and CA was the disabling of pneumatic valves on this vehicle.

III.4. Functioning of buildings and devices intended for railway traffic and railway vehicles

III.4.4. Functioning of railway vehicles, including analysis of entries from on-board data recorders

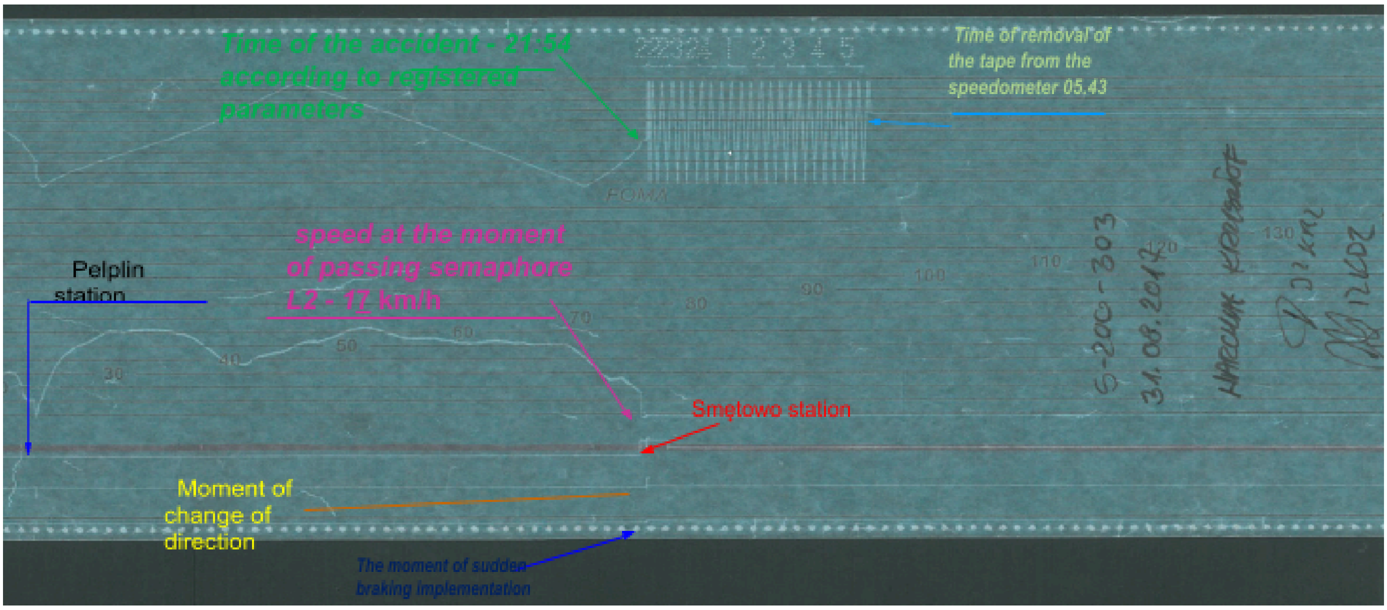
TMS train 564024 from Gdynia Port GPA - Wrocław Gądów.
The train drive with the consent ID No. 5-6220-466_2016 and in accordance with the individual timetable generated in the SKRJ system on 29.08.2017 at 17:22.
Train was composed of a locomotive and 6 cars with a total length of 176 m and a weight of 401 tons.
Locomotive S200-303 EVN no. 925136200761.
Certificate of technical efficiency STK / 01/2017 year of construction 1981; serial number 11314; produced by CKD PRAHA, for which a certificate of release for exploitation of type of railway vehicle No. T / 2003/0095 was issued; having the identifier of the railway vehicle PL-ID 925136620076-1; certificate valid until 30.01.2023; mileage 600,000 km,
counted from 230 km Rybnik 01/02/2017. Transit and acceptance protocol of the S200-303 locomotive after the inspection repair (P4) on 31.01.2017.
The locomotive in the non-derailed condition - the vehicle's front at km 457,447.
cars in the following order after the locomotive in a non-derailed state:
PL-STK 605199080017 series: Sk,
PL-STK 335199330085 series: Rs,
PL-STK 845139360565 series: Rs,
PL-STK 335149403130 series: Uaai type: Krupp 32,
PL-STK 375597000021 series: Sr,
PL-STK 838199733208 series: Uaai type: Krupp 32.
The brake test card and the list of vehicles in the TMS \ 564024 train were located in one of the cars of train TMS \ 564024 - PL-STK 605199080017 series Sk.

The train was controlled from the driver's desk "A". Siren operation correct. According to the expertise of the radiotelephone manufacturer,
it was not possible to clearly state the channel number on which the device worked at the time of the accident.

Signaling the front and end of the train - signals "Pc 1" and "Pc 5", in accordance with the instruction Ie-1 (E-1).

Description of the data from the speedometer recording tape type Hasler BERN Rt9 No. M08.080 of the S200-303 series traction unit:

The description is related to the event that took place on 30.08.2017 at 21:53 in Smętowo station at km 457.48, line No. 131: Chorzów Batory - Tczew, while driving train 564024 Gdynia Port GPA - Wrocław Gądów.



Vehicle: The locomotive S200-303,
Hasler BERN RT9 type recorder no. M08.080,
The company's speedometer tape: FOMA,
Measuring range of the tape: 150km/h.
The analysis concerns the range from 21:30 to 05:43, i.e. the time of removing the tape from the recorder.
The following parameters are recorded on the tape:

- graph of the time function,
- graph of the speed function,
- traveled way,
- brake cylinder pressure - braking.

No registration of the vigilance signals on the speedometer tape
This is the Automatic Brake for the Train and Deadman's Handle (SHP and CA) and the lack of vehicle registration with the drive switched on and off - current and dry running.

21:27 - recorded pressure in the brake cylinders and temporary stop of the train at the Pelplin station,
21:30 - there is an increase in speed on the road of 2.5 km to the speed of about 62 km / h and driving at this speed on the section of 2.2 km,
21:35 - there is a slow drop in the speed of the train without the pressure increase in the brake cylinders on the road 2.5 km to the speed of 40km / h and then the train speed increase to 65km / h on the road 4.2 km and again the free fall of the speed to 55km / h on 1.3 km and the train running at a speed in the range from 55 to 50 km / h on the road 5.5 km to 21:49,
21:49 - at this time there is a free fall in speed from 50km / h to 27km / h on the section 1.3 km, then driving at a speed of 26-28 km / h on a section of about 0.5 km,
21:52 - another free fall in speed on the road about 700m to about 17km / h,
21:54 - a rapid drop in speed to 0km / h. Before the speed drops dramatically, directional change is switched on without braking. At the moment of impact, the recorded vibrations of the stylus on the tape and after stopping recorded the pressure increase in the brake cylinders and

the application of the emergency braking of the train. The train stops from 21:54 to 05:43 on 31.08.2017 and removes the tape from the recorder by the members of the railway commission.

The SHP and CA devices on the locomotive are turned off.

During the inspection at the place of the event and on the basis of data from the speedometer recorder of the type Hasler BERN Rt9 No. M08.080, vehicle S200-303, **the research team concluded that the clearing signals of the SHP and CA alert devices were not recorded due to the valves being moved to the closed position. However, the driver's use of "emergency braking" was recorded on the speedometer tape immediately after the incident.**

FOMA speedometer tape with speed range up to 150km / h is not recommended
by the manufacturer of speedometers (Hasler BERN) for use.

The time indicated in the recorder type **BERN** type **Rt9**has been adopted by the research team as the basis for analyzes aimed at determining the circumstances and causes of the event.

Carrier STK S.A. delivered on 24 July 2018 a printout showing the recording of the GPS position monitoring system of the locomotive S200-303, from which it follows that immediately before the accident for 5 minutes the parameters of the locomotive engine's operation did not change, which confirms the lack of response to signal indications at semaphores.

IV.1. Reference to earlier accidents or incidents occurring in similar circumstances

In three years before a serious accident occurred, an accident occurred in similar circumstances - cat. B04 on 16.03.2017 at 4:55 at the Poznań Franowo PFB station. The Freight Train No. 780001 carrier PKP Cargo S.A, leaving the track No. 314 was driven into by a SM42-1230 locomotive pulling 31 cars from
maneuvering track no. 214. The driver of the maneuvering set-up started the depot from under the maneuvering disk Tm 220 indicating the signal Ms1 "maneuvering forbidden" and continued the ride of this depot, which caused a clash with the outgoing train. As a result of a later clash,
5 cars of the freight train derailed and the SM42 shunting locomotive and the railway surface were damaged. The railway commission conducting the proceedings formulated two preventive applications:
- for the carrier - regarding additional maneuvering team responsibilities and
- for the manager and carrier regarding the elimination of irregularities in the service of radiotelephones and the transmission of radio messages by employees.

IV.3. Findings of the research team regarding the course of the accident based on existing facts

The research team of PKBWK determined that the train driver of the MTS 564024 train, driving the railway vehicle at 17 km / h with the vigilance devices turned off, did not notice the broadcast signal S1 "Stop" at the L^2 signpost semaphore for track No. 32
and he did not implement the braking, which led to the omission of the semaphore and entering the track
No. 2, occupied by the MPE 54170 passenger train. The research team of PKBWK sees the cause of the occurrence on the side of the train driver of the MTS 564024 freight train.

IV.6. Indication of other irregularities revealed during the proceedings, but not relevant to accident claims

Other irregularities identified in the proceedings, which do not have a direct impact on the occurrence of the event, include:

- Lack of implementation by STK SA employees. the provisions of "P04 transport procedure" of the Safety Management System regarding the completion of the "List of railway vehicles in the train set". The list of railway vehicles in the train train for TMS 564024 has been drawn up by an unauthorized person - the rolling stock auditor,
  and it should be made by the train driver, as provided in the entry *"P04 transport procedure"* in the scope of tasks for the traction team.
- There are no cameras recording the route image on the STK SA carrier vehicles, despite the fact that the President of UTK issued a resolution No. DBK-550 / R-03 / KB / 12 of 30/05/2012, addressed to rail operators with the obligation to install recording devices - digital cameras or video recorders in railway vehicles newly built and in operation in accordance with the recommendation of GDPWK - No. PKBWK-076-305 / RL / R / 11 of 22.11.2011.
- No registration on the speedometer tape from the S200-303 locomotive parameters concerning the operation of train safety devices, i.e .: SHP and CA while driving a TMS train No. 654001 between Czersk - Gdynia GPA and TMS 564024 train.
- Driving of the locomotive S200-303 leading the train on the day preceding accident No. 654001/0 of Czersk - Gdynia Główna on 29.08.2017 and on the day of the accident train number 564024 with the speed of decommissioning with devices of train safety devices off (SHP and CA), which is inconsistent with the provisions of § 63 para. 13 Instruction Ir-1 (R-1) version from 2015 - text unified from 09/05/2017 "P04 transport procedure" in the scope of tasks specified for the traction team.
- Lack of supervision over the speedometer tape management by the STK S.A. which is inconsistent with the provisions of the "*P04 transport procedure*" as well as § 19 paragraph 1 point 6 and 9 of the *STK-1 Instruction for driver and helper of diesel locomotive and*
  electric company STK S.A.
- Use of speedometer tapes not recommended by the manufacturer .
- Lack of the brake test card for this train on the locomotive running TMS 564024, which violates the provisions of § 7 para. 4 point 5 and § 13 para. 2 of the *STK-1 Instruction for the train driver and helper of the* STK SA locomotive and electric locomotive and § 4 para. 1 of the *Regulation of the Minister of Transport of*
  2.11.2006 on documents that should be in a railway vehicle.
- After completion of train no. 654001/0 Czersk - Gdynia Główna on 29.08.2017, no drivers STK S.A. removing the part of the tape from the speedometer
  of the S200-303 locomotive and its description, which violates the provisions of § 19 para. 1 point 6 of the *STK-1 Instruction for the driver and helper of the STK S.A. locomotive of the diesel and electric locomotive*. According to the research team, parts of the tape were not removed and not described as an internal rule, because the same driver was driving another train number 564024 with the same locomotive the following day
  i.e. 30.08.2017 using the same tape.

- Incorrect performance of the P1 level review of the S200-303 locomotive by the employee of the Maxima Joint-Stock Company in Gdańsk - which results from the scope of activities included in sheet No. 7, S200 Maintenance System Documents - no verification of the correctness of recording on the vigilance tape of the alert and deadman's handles and assumption or filling the seals, as demonstrated in point II.1.4.
- Lack of checking by the driver of the technical condition of the traction vehicle and vigilance devices when starting work on the vehicle, which is inconsistent with the provisions § 11 para. 4 point 6e of the STK-1 Instruction for train driver and helper of the STK SA locomotive and electric locomotive..
- Failure by drivers of STK S.A. to maje an annotation in "Drive Vehicle Book" regarding the travel of a superior officer and another driver in the driver's cab of the S200-303 vehicle while running the train no. 654001/0 from Czersk to Gdynia Główna on 29.08.2017, which is a violation of the provisions of § 9 para. 6 of the STK-1 Instruction for driver and helper of the *STK SA locomotive and electric locomotive.*.
- Incorrect way of keeping documentation regarding the acquisition of knowledge of sections of railway lines by train driver 564024, consisting in entering in the route knowledge of the date of purchase, despite only dark driving, which violates the provisions of § 10 para. 5 of the *STK-1 instruction for the driver and helper of the engine locomotive of STK S.A.* and Procedures P / 12 - "*Personnel competence management".*
- Lack of the current date and time of the manipulator 015 / OG RTC clock and performance of mandatory inspections of the train radio devices set F-747 Pyrylandia located in the locomotive S200-303, which violates the Instruction on principles of performing technical service of rail telecommunications devices - Ie-13 PKP PLK SA
- Employing a train driver under contract, which was inconsistent with the rules in force at STK S.A. - Procedure P / 12 - "Personnel competence management" orders "signing a contract of employment", not contract of mandate.
- Built-in indicator W3 on the right side of track No. 2 at kilometer 457,486 at which there is no built-in semaphore, can not be justified in the current internal regulations of the infrastructure manager, § 16 para. 15 point 3.
- Incorrectly formulated announcement telephones on the announcement link and radiographs of Smętowo station not in accordance with the provisions of § 24 para. 18 of Ir-1 Manual as well as § 39 section 7.
- Removal of TMS 564024 from the Gdynia Port GPA start station inconsistently with an individual timetable for this train.

I. RECOMMENDATIONS TO AVOID SUCH ACCIDENTS IN THE FUTURE OR LIMIT THEIR EFFECTS

The PKBWK research team recommends the implementation of the following activities:

1. STK S.A. shall execute the order of the President of the Office of Rail Transport No. DBK-550/R-03/KB/12 of 30.05.2012, addressed to railway carriers with the obligation to install recording devices - digital cameras or video recorders in newly built and in operation railway

vehicles , in accordance with the recommendation of PKBWK – Nr PKBWK-076-305/RL/R/11 of 22.11.2011.

2. As part of the safety management system, STK S.A.
   a. will analyze and evaluate the risk for cases of running railway traction vehicles with the intention of shutting down the SHP and CA alertness devices and not complying with the provisions of § 63 para. 13 of Instructions Ir-1 (R-1),
   b. will increase as part of the safety improvement program for the following years: the number of speedometer tape controls and the number of control trips in the traction vehicle cabins,
   c. will increase the number of security audits in the framework of the security improvement program for the coming years, in particular with regard to the transport process, with particular emphasis on supervision of the work of drivers and rolling stock auditors.
3. STK SA will increase supervision over the work of train drivers in terms of railway safety, and especially driving vehicles with SHP and CA devices turned on.
4. PKP PLK S.A. as part of periodic instructions for directly related persons with the railway traffic will put a special emphasis on:
   a. the need for employees to check traffic posts before giving the permission signal on the semaphore, whether the route is prepared, i.e. whether it is properly set and secured and there are no obstacles to drive ", which is required by § 40 para. 1, 2 and 3 of the Ir-1 instruction,
   b. correct formulation of telephoneograms and radiographs announcers,
   c. good practice of informing by radio the drivers of railway vehicles driving by traffic police officers about changes in the traffic organization of a given train within the station, in particular about unplanned stops at the station, in order to pass other trains.
5. STK S.A. verify the correctness of the inspections performed:
   a. train radio equipment operated by subcontractors and increase the supervision over the performance of these services,
   b. traction vehicles in order to comply with the provisions of the Maintenance System Documentation of this series of vehicles and increase the supervision over the performance of these services.
6. STK S.A. will take actions to supervise the train radio devices in terms of compliance of the time of these devices with real time.
7. The location of the "W3" indicator at track No. 2 at kilometer 457,486 at Smętowo station is not justified by the current internal regulations. PKP PLK S.A. Zakład Linii Kolejowych in Gdynia, at the request of the PKBWK, he liquidated this indicator during the proceedings.
8. Infrastructure manager of PKP PLK S.A. will take measures to adapt the internal rule Ie-4 (WTB-E10) in the scope of protective routes taking into account train journeys by the station at different speeds, when these trains enter the route equipped with in a multi-block automatic line lock.
9. To obtain an effective protection route of 50 m to the turnoff of the Rz24 and Rz25 crossroads after the L^2 and K^2 signposting semaphores with the speed of trains passing by 120 km / h through the station on the main main track, the manager will consider the possibility of moving the semaphores to the Sm2 control room by a minimum of 28 meters, or, without moving the semaphores, build gauge blocks at the extension of track No. 32 and 31 at least 50m behind the^2 i K^2. semaphores. The current application of the protection route is in accordance with the applicable internal regulations of the Ie-4 manager (WTB-E10).

As part of the system approach to safety, railway infrastructure managers will analyze protective routes used at traffic posts with similar conditions of traffic organization, including the track system.

10. Infrastructure manager PKP PLK S.A.. until the protection route for the course of z^2_{32} and z^2_{31} before the intersections of switches 24 and 25, will be introduced in the Technical Standpoint of the Movement Post (RTPR) - station Smętowo:
    a. before proceeding with the preparation of the route for the train entry to track No. 2 from the Z-semaphore for route $Z^1_{2,}$, the duty of the train operator to make sure that the train stops after completing the course on track No. 32,
    b. before proceeding with the preparation of the route for the train entry to track No. 1 from semaphore A for route A^1, the duty of the train operator to make sure that the train stops after completing the course on track No. 31.

According to art. 28l par. 8 of the Act of 28 March 2003 on railway transport (Dz. U. of 17. item 2117, as amended), the above recommendations are directed to the President of the Office of Rail Transport, who exercises statutory supervision over infrastructure managers and carriers. Individual entities should implement the recommendations contained in this Report of the Research Team and adopted by the resolution of the GDPWK.